

WINKLER COUNTY AP, TX, USA

WMO: 722656

Lat: 31.780N

Lon: 103.202W

Elev: 2807

StdP: 13.27

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 23040

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.6	24.5	-0.6	5.9	49.2	3.0	7.1	49.9	31.1	60.1	26.9	60.6	5.6	350	0.535

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.2	105.5	66.4	102.5	66.7	100.0	66.8	72.9	89.3	72.0	88.6	71.2	87.9	10.1	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.6	120.8	76.3	68.2	114.9	76.1	66.9	110.0	76.3	38.4	89.2	37.5	88.3	36.7	87.5	78.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
26.0	22.2	19.2	13.7	110.0	4.2	2.7	10.7	111.9	8.2	113.5	5.9	115.0	2.8	116.9	
			11.3	75.1	3.9	1.4	8.5	76.2	6.3	77.0	4.1	77.8	1.3	78.8	
			DB												
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	66.5	46.0	50.8	58.5	67.1	76.0	84.2	85.1	84.3	77.2	66.7	54.6	46.1
	DBStd	16.14	8.39	8.99	8.78	8.01	7.37	5.08	4.20	4.26	6.13	8.16	8.43	8.10
	HDD50	526	178	91	29	2	0	0	0	0	0	4	50	172
	HDD65	2296	591	400	233	71	11	0	0	0	4	77	321	588
	CDD50	6534	53	113	292	514	807	1027	1089	1062	817	522	188	50
	CDD65	2827	0	2	31	133	353	577	624	597	371	130	9	0
	CDH74	37942	59	205	820	2213	4791	7795	8316	7640	4141	1610	297	55
	CDH80	21365	6	45	286	1060	2754	4800	5003	4507	2158	673	69	4
Wind	WSAvg	9.3	8.2	9.2	10.1	10.9	10.4	10.8	9.7	8.9	8.7	8.7	8.1	8.1
Precipitation	PrecAvg	11.5	0.4	0.5	0.5	0.6	0.9	1.6	1.8	1.3	2.0	1.2	0.4	0.4
	PrecMax	22.6	1.5	1.5	3.0	4.8	8.7	4.8	5.9	3.1	9.1	5.6	2.4	2.6
	PrecMin	4.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0
	PrecStd	4.0	0.4	0.4	0.7	0.9	1.5	1.3	1.7	0.9	2.3	1.5	0.6	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.9	85.4	91.3	97.7	106.8	109.6	107.3	106.3	102.2	95.4	86.9	79.5
		MCWB	52.0	53.7	56.4	59.5	63.5	65.0	68.4	69.1	66.9	64.1	58.1	52.4
	2%	DB	74.8	80.2	87.2	94.0	101.8	106.1	103.5	102.3	98.0	91.1	81.3	74.5
		MCWB	50.6	52.6	55.6	58.0	62.1	65.6	68.1	68.6	66.5	62.9	55.8	51.5
	5%	DB	69.8	75.4	83.0	90.4	98.2	102.7	100.8	99.8	94.9	87.7	77.2	68.6
		MCWB	48.5	50.9	54.3	57.2	61.6	65.9	68.4	68.5	66.7	62.0	54.7	49.1
	10%	DB	64.1	70.2	78.8	86.7	94.1	99.3	98.6	97.3	91.4	83.4	72.6	63.5
		MCWB	46.4	49.1	53.2	56.4	60.8	66.5	68.4	68.6	66.6	61.0	53.6	47.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.3	58.6	64.0	67.3	71.4	74.3	74.4	73.7	72.5	71.0	64.3	57.4
		MCDB	67.9	70.0	76.9	81.6	86.9	90.0	89.3	91.0	86.9	83.8	72.8	66.5
	2%	WB	52.9	55.6	60.8	64.9	69.4	72.6	72.7	72.5	71.3	68.9	60.3	53.5
		MCDB	67.8	70.6	73.0	78.1	84.1	89.5	88.6	89.7	85.4	81.0	71.1	67.7
	5%	WB	50.2	53.2	58.4	62.7	68.1	71.4	71.7	71.5	70.4	67.1	58.2	51.0
		MCDB	66.7	70.1	72.4	77.4	83.0	88.1	88.6	88.9	84.5	78.1	69.5	66.6
	10%	WB	47.5	50.8	55.7	60.4	66.5	70.4	70.9	70.9	69.4	65.0	55.8	48.2
		MCDB	63.4	68.5	72.7	75.7	82.2	87.5	88.3	88.1	83.4	75.8	69.6	62.5
Mean Daily Temperature Range	5% DB	MDBR	30.2	31.0	31.4	32.3	30.1	27.9	25.2	24.9	25.0	28.0	29.7	29.3
		MCDBR	40.5	40.6	39.4	37.2	36.1	32.8	28.8	27.9	29.9	33.8	37.5	39.3
		MCWBR	20.7	19.3	15.9	14.3	12.0	8.7	7.3	7.0	8.4	11.9	16.9	20.3
	5% WB	MCDBR	35.0	34.2	31.4	30.3	28.7	26.8	24.4	24.3	23.2	23.4	28.1	33.9
		MCWBR	18.3	17.1	14.6	13.6	11.7	8.2	6.6	6.4	7.2	9.3	13.6	18.4
Clear-Sky Solar Irradiance	taub	0.261	0.269	0.287	0.324	0.342	0.375	0.397	0.371	0.354	0.305	0.282	0.266	
	taud	2.585	2.565	2.520	2.398	2.410	2.371	2.350	2.405	2.425	2.557	2.552	2.568	
	Ebn at Noon	312	318	317	306	299	288	281	288	290	300	300	304	
	Edh at Noon	25	28	32	38	38	39	40	37	35	28	26	24	
All-Sky Solar Radiation	RadAvg	1120	1398	1803	2196	2344	2421	2294	2127	1797	1484	1177	1004	
	RadStd	103	134	124	131	143	110	110	95	118	144	113	85	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page